Distillation

Pre-Lab Plan
Unit Operations Lab #4
Date 10/15/19

Group #4, Tuesday, 9 AM section
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1. Experimental Objective (5 pts)

The objective of the lab is to vary the boiler power and reflux ratio, and determine the pressure of the column, the boil up rates, the column efficiency, and the product composition at each variation.

2. Experimental

2.1 Apparatus (5 pts)

Figure 1, shows the apparatus that will be used in the lab.

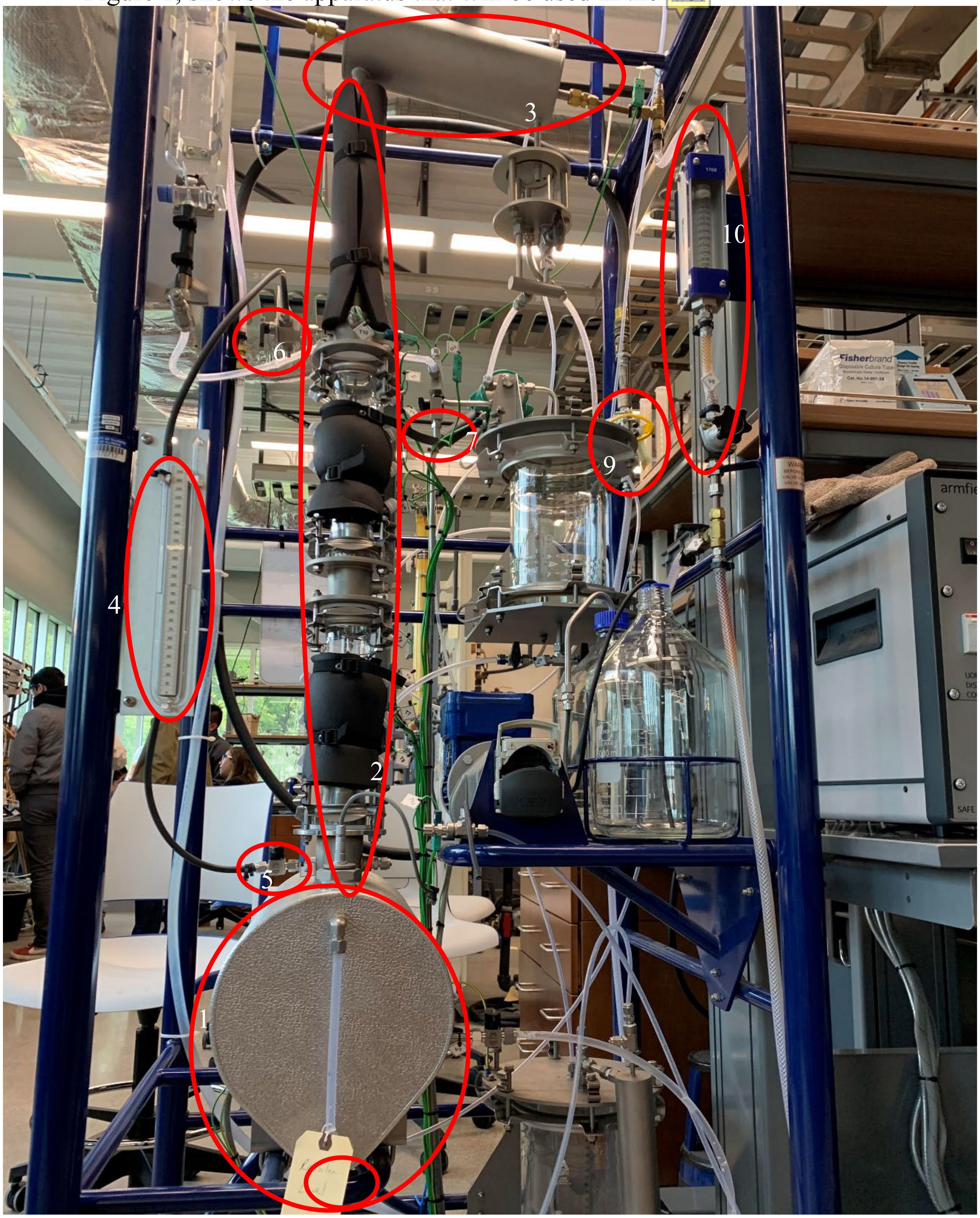

Figure 1: Apparatus

Below in Figure 2 is the PFD of the apparatus supplied by armfield.

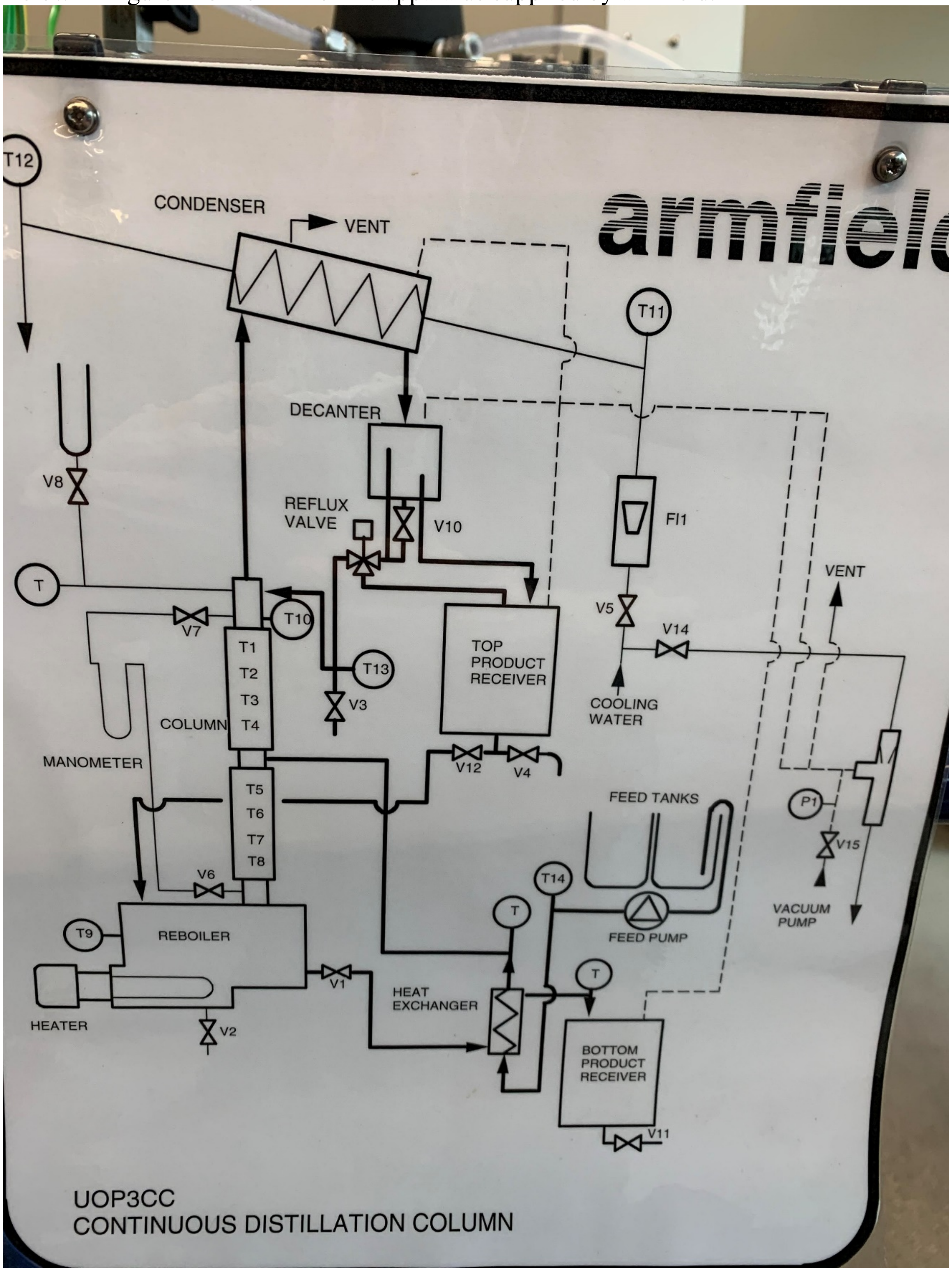

Figure 2: PFD

Table 1 is the list that describes each of the pinpointed spots on our apparatus.

Component	Description
1) Reboiler	Supply heat to liquid mixture
2) Column	Liquid rises through column
3) Condenser	Liquid is condensed into distillate
4) Manometer	Differential Pressure reading from column top and reboiler
5) Pressure Valve	High Pressure valve connected to the reboiler
6) Pressure Valve	Low pressure valve connected to the top of the column
7) Ball Valve	Collection point from the condenser
8) Ball Valve	Collection point from the bottom of the reboiler
9) Ball Valve	Allows flow of cold water into system
10) Needle Valve	Controls amount of water flow allowed into system

Table 1: Apparatus component list

2.2 Materials and Supplies(5 pts)

Table 2 describes the material and supplies that will be used in the lab.

Materials/Supplies	Description
1) Ethanol	200 proof pure denatured ethanol used to create solution
2) DI water	Mixes with ethanol to create solution
3) Refractometer	Measure impurities in our fluid based off refractive index
4) Test tubes	Collects fluid from collection points in apparatus in section 2.1
5) Kimwipes	Cleans refractometer without scratching it
6) Cold water bath	Cools down the liquid collected from the reboiler
7) Heat resistant gloves	Protects from heat when collecting the hot fluid from reboiler

2.3 Experimental Plan (10 pts)

In this lab a Distillation Column is used for separating a binary mixture of 10% weight ethanol and water. The apparatus consists of a floor-standing process unit (column, condenser, and reboiler) and a bench-mounted control console. A PC computer is available with the Armfield software and is connected to the console for operating the column. The feed for analysis enters near the middle of the column above the reboiler. Heat is supplied to the reboiler to boil the mixture. Separated vapor travels up the column and liquid travels down the column in a counter-current flow pattern. Two experiments will be conducted.

Preparation:

1. To begin, make up 10 liters of a mixture of 10% weight ethanol and water and set the equipment to operate at total reflux by using the PC console and turning off the reflux ratio timer.
2. After ensuring all valves of the equipment are closed, open valve V10 on the reflux pipe.
3. Fill the column with the 10 liter ethanol/water mixture through the filler cap on top of the reboiler by using a funnel/blue hose. Secure the filler cap after filling is complete.
4. The 1st and 5th plates in the column are unobscured and can be observed during the distillation process.

Experiment A: Effect of Boil-Up Rate and Overall Column Efficiency

The objective of this experiment is to determine the variation with boil-up rate of pressure drop over the distillation column and to determine the overall column efficiency. To accomplish this, the amount of heat added to the system via the reboiler (measured in kW) will be varied and the effect that has on boil-up rate and the pressure drop across the vessel will be observed.

Procedure:

1. Once this preparation is complete, turn on the control panel and set the temperature selector switch to T9, which sets the temperature to the reboiler.

1. Open the valve V5 to set the cooling water flow rate to approximately 3 liters/min.
2. On the control panel, turn the power controller to 1.50 kW and turn the switch for powering the heat element.
3. Open valves V6 and V7 which connect the base and top of the distillation column to the manometer creating a pressure difference. Now close valve V6 and V7.
4. Vapor will start to rise up the column as temperature is increased by the temperature selector for T8, T7, T6, T5, T4, T3, T2, and T1.
5. The system is at equilibrium at given power rates when the selected temperatures are constant.
6. Vapor will enter the condenser and distillate will collect in the receiver and eventually overflow into the reflux regulator valve.
7. Once there is steady bubbling on all trays, the power controller is turned clockwise to 0.50 kW and the cool distillate will travel back down the trays.
8. The boil-up rate at 0.50 kW is measured by opening valve V3 and using a stopwatch to measure the time required for the condensate to drain from the reflux system and collect into a measuring cylinder.
9. Measure the refractive index of the overhead samples and measure the composition of the samples using the Refractometer.
10. Use the same procedure to collect a sample of the bottom through the valve V2.
11. Measure the refractive index of the bottom sample also.

12. After samples are collected, observe pressure drop for both the top and bottom sections by opening the valves V6 and V7 on the manometer. Close the valves when done with the reading.
13. Repeat experiment A until at least 3 samples from the overhead and bottom are collected.
14. Record the temperatures of T8 and T1 for each experiment to calculate an average column temperature.

Experiment B: Distillation at Constant Reflux Ratio

The objective of this experiment is to vary reflux ratios, therefore varying the top and bottom compositions with time.

<u>Procedure:</u>

1. Use the same ethanol/water mixture from Experiment A which is still at high temperature.
2. Set the heat controller (giving a total reflux of about 0.75 kW) and leave the apparatus for about 15 minutes to let the system reach equilibrium state.
3. Set the reflux controller to the first desired value by setting 20 seconds back to column and 4 seconds to top product receiver. Once this is done, use the control panel to switch the reflux valve on.
4. Similar to experiment A, measure the boil-up rate with a timed volume collection by opening valve V3. The difference is a small discarding sample of 5 to 10 mL is collected in a separate vessel and the remaining is diverted to the measuring cylinder. Time is measured for collection of the set quantity. Collect a sample three times and take an average value for the boil-up rate.
5. Repeat every ten minutes to obtain 5 samples of both the overhead and bottom.
6. Repeat experiment B with at least one other reflux ratio.
2. Record the temperatures of T8 and T1 for each experiment to calculate an average column temperature.

Independent Variable: Pressure, Temperature
Dependent Variable: boil-up rate, overhead composition, bottom composition
Specification Ranges: Temperatures @ T8, T7, T6, T5, T4, T3, T2, and T1

Statistical and/or calibration methods used to estimate/reduce experimental uncertainty:

Subscripts d, b indicate distillate and bottom. The efficiency is given by:

Knowing the composition of distillate and bottom and the corresponding volatilities, the column efficiency can be determined:

Confidence Interval: 85% confidence interval. PC equipment will be used but will have human error with collecting samples and measuring time with the stopwatch.

Project Safety Evaluation (10 pts)

Project Title: Binary Distillation

Date: 10/15/19

	Yes/No	
Potential electrical hazards?	Yes	There is a significant amount of electrical connections between the process unit and the control console. Each of these connections along with the power source for the control console represent a risk of electric shock.
Potential mechanical hazards?	Yes	There are a number of mechanically operated pieces (such as valves and other flow control equipment) which must be manually operated, and each of these represent a potential risk of mechanical failure.
Potential pressure hazards?	Yes	As there is no gauge attached to the system that allows for comparison of column pressure to atmospheric pressure, an absolute or relative pressure for the column cannot be given. However, we can say that the column will be operating at a pressure slightly higher than atmospheric and that the group will monitor for leaks and blockages of flow. This risk can be categorized as minor.
Potential temperature hazards?	Yes	The reboiler attached to the operating unit is capable of heating mixture temperature to almost 100 C and represents a significant burn risk. Thermal Shielding Gloves will be used for any operation in the vicinity of the reboiler, particularly when taking liquid samples of the bottom's mixture.
Hazardous/toxic chemicals involved?	Yes	Denatured ethanol is specifically formulated to make it unfit for human consumption. See the MSDS sheet for Ethyl Alcohol.
Gloves required?	Yes	Gloves should be worn for all parts of the experiment. Heat Resistant gloves when in the vicinity of the reboiler and nitrile gloves everywhere else.
MSDS reviewed by all team members?		List MSDS sheets reviewed: Ethyl Alcohol, Denatured (ACC# 08701)
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	100 ºC	Maximum temperature of the reboiler would be the boiling point of our mixture. However, since water is used technically the boiling point of water would be the maximum temperature possible.
Pressure (psia)	Unknown	The max pressure in the lab would be the maximum pressure inside the column. This pressure itself would not be very high, can be assumed to be slightly above atmospheric.
Safety Controls	*yes/no*	*If yes, Provide description*
	no	

I have read relevant background material for the Unit Operations Laboratory entitled: "Binary Distillation" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Michael Stacy	10/15/19
signature of team member	date
Emir Asani	10/15/19
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